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program SOR_Method
  implicit none
  integer :: i, j, k, n, M
  real(8) :: omega, TOL, sum1, sum2, diff, norm
  real(8), dimension(6,6) :: A
  real(8), dimension(6) :: b, x, X0

  ! Initialize parameters
  n = 6
  omega = 1.1d0
  TOL = 1.0d-5
  M = 100
  x = 0.0d0
  X0 = 0.0d0

  ! Coefficient matrix A
  A = reshape( [ &
    4.0d0, -1.0d0, 0.0d0, 0.0d0, 0.0d0, 0.0d0, &
    -1.0d0, 4.0d0, -1.0d0, 0.0d0, 0.0d0, 0.0d0, &
    0.0d0, -1.0d0, 4.0d0, -1.0d0, 0.0d0, 0.0d0, &
    0.0d0, 0.0d0, -1.0d0, 4.0d0, -1.0d0, 0.0d0, &
    0.0d0, 0.0d0, 0.0d0, -1.0d0, 4.0d0, -1.0d0, &
    0.0d0, 0.0d0, 0.0d0, 0.0d0, -1.0d0, 4.0d0 ], &
    shape = (/6,6/) )

  ! Right-hand side vector b
  b = [3.0d0, 3.0d0, 3.0d0, 2.0d0, 2.0d0, 3.0d0]

  ! SOR Iteration
  do k = 1, M
    norm = 0.0d0
    do i = 1, n
      sum1 = 0.0d0
      sum2 = 0.0d0
      do j = 1, i-1
        sum1 = sum1 + A(i,j) * x(j)
      end do
      do j = i+1, n
        sum2 = sum2 + A(i,j) * X0(j)
      end do
      x(i) = (1.0d0 - omega) * X0(i) + (omega / A(i,i)) * (b(i) - sum1 - sum2)
    end do

    ! Compute norm ||x - X0||?
    do i = 1, n
      diff = abs(x(i) - X0(i))
      if (diff > norm) norm = diff
    end do

    if (norm < TOL) then
      print *, 'Converged after', k, 'iterations.'
      exit
    end if

    X0 = x
  end do

  ! Print result
  print *, 'Solution vector x:'
  do i = 1, n
    print *, 'x(', i, ') = ', x(i)
  end do
end program SOR_Method

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